



Module 2: Operating & Propagation

Repeaters · Propagation · Station Setup
2026–2030 Technician Question Pool



Part 1: Station Operation Basics

Subelement T2A · 1 exam question



Before You Transmit

- LISTEN FIRST — always
- Then ask: 'Is this frequency in use?'
- Wait for a response before transmitting
- This is the most basic operating courtesy
- Applies every single time you go to transmit
- Skipping this step = instant bad reputation



Making a Call

- General call (CQ): 'CQ CQ CQ, this is [your call], standing by'
- Calling a specific station: '[their call], this is [your call], over'
- Responding to CQ: '[their call], this is [your call], over'
- Ending: '[their call], this is [your call], 73, clear'
- 73 = 'best regards' — the universal ham goodbye
- Always end with your call sign (it's your final ID)



Repeater Offsets

- A repeater listens on one frequency, transmits on another
- The difference = the offset
- 2 meters: ± 600 kHz (0.6 MHz)
- 70 cm: ± 5 MHz
- 6 meters: ± 1 MHz
- Your radio applies the offset automatically
- You just need to know + or – for each repeater



Part 2: VHF/UHF Practices

Subelement T2B · 1 exam question



CTCSS (PL Tones)

- Sub-audible tone your radio sends to 'unlock' the repeater
- Without the correct tone, the repeater ignores you
- Each repeater publishes its required tone
- Also called PL tone (Motorola trade name)
- Set it in your radio BEFORE you try to talk
- Common exam question: what is CTCSS used for?



DMR Basics

- Digital Mobile Radio — digital voice mode
- Uses time slots (2 conversations per frequency)
- Uses talk groups (virtual channels)
- Requires a free DMR ID from radioid.net
- Color code = DMR version of a CTCSS tone — must match repeater
- Growing fast — many repeaters now support DMR



Q Signals

- QRM — man-made interference
- QRN — natural noise/static
- QSY — change frequency
- QSL — confirmed/acknowledged
- QRZ — who is calling me?
- QTH — my location is...
- These are used on both phone and CW



Part 3: Emergency Comms

Subelement T2C · 1 exam question



ARES vs. RACES

ARES (Amateur Radio Emergency Service)

- Run by ARRL (volunteer)
- FCC license required
- Activates anytime by Emergency Coordinator
- Broader scope — any emergency
- Most active emergency comms organization

RACES (Radio Amateur Civil Emergency Service)

- Run by local/state civil defense (government)
- FCC license AND civil defense enrollment
- Only during declared emergencies
- More restricted scope
- Often overlaps with ARES membership



Emergency Communications

- In a genuine emergency: use ANY frequency needed
- Normal FCC rules bend to save lives
- This is explicit in Part 97 — not a gray area
- Net operations: net control directs all traffic
- Tactical call signs OK during emergencies (still ID every 10 min)
- ARES volunteers are the backbone of ham radio public service



Part 4: Radio Waves

Subelements T3A & T3B · 2 exam questions



The Wavelength Formula

- Radio waves are electromagnetic waves — travel at the speed of light
- Wavelength (meters) = $300 \div \text{Frequency (MHz)}$
- $146 \text{ MHz} \rightarrow 300 \div 146 \approx 2 \text{ meters}$ (the '2-meter band')
- $440 \text{ MHz} \rightarrow 300 \div 440 \approx 0.7 \text{ m} = 70 \text{ cm}$ (the '70-cm band')
- $28 \text{ MHz} \rightarrow 300 \div 28 \approx 10.7 \text{ m}$ (the '10-meter band')
- This is the ONE formula you absolutely must know



Polarization & Propagation Basics

- Vertical antenna → vertically polarized wave
- Horizontal antenna → horizontally polarized wave
- Mismatch = significant signal loss (cross-polarization)
- VHF/UHF: mostly line of sight — blocked by terrain and earth's curve
- HF: bounces off the ionosphere — can travel worldwide
- Multipath: VHF signals reflecting off buildings → flutter/fading



Part 5: Propagation Modes

Subelement T3C · 1 exam question



Propagation Modes

- Line of sight — VHF/UHF normal; limited by terrain and earth's curve
- Ground wave — follows earth's surface; low HF, AM broadcast
- Sky wave (F-layer skip) — HF bounces off ionosphere → worldwide
- Sporadic E — surprise VHF skip; 6m, sometimes 2m; 1000+ miles
- Tropospheric ducting — weather bends VHF/UHF; extended range
- Meteor scatter — brief bounces off meteor trails; 6m, 2m
- Auroral — reflects off aurora; distorted signal



Part 6: Station Setup

Subelements T4A & T4B · 2 exam questions



Essential Station Controls

- Volume — audio level (doesn't affect transmitted signal)
- Squelch — mutes receiver until signal exceeds threshold
- Frequency/VFO — sets your operating frequency
- Offset — sets repeater offset (+ or −)
- CTCSS/Tone — sets the sub-audible tone
- Power — adjusts transmit power level
- PTT — push to talk; hold to transmit, release to listen
- AGC — automatic gain control; keeps audio levels steady



Microphone Technique

- Hold 2–3 inches from mouth, slightly to the side
- Normal speaking voice — don't shout into the mic
- Wait a beat after pressing PTT before speaking
- Keep transmissions short — say it, then unkey
- Don't 'double' — wait for the courtesy tone/tail
- Listen more than you talk, especially when you're new



Key Numbers for the Exam

Operating

- 2m offset: ± 600 kHz
- 70cm offset: ± 5 MHz
- 2m calling: 146.520 MHz
- CTCSS: sub-audible tone for repeater access
- DMR color code: like CTCSS for digital
- Emergency: ANY frequency allowed
- QRM = man-made, QRN = natural

Propagation & Waves

- Wavelength = $300 \div \text{MHz}$
- VHF/UHF: line of sight
- HF: sky wave (ionosphere)
- Sporadic E: surprise VHF skip
- Cross-polarization = signal loss
- Squelch: mutes until signal present
- AGC: keeps audio levels steady



Checkpoint Quiz

10 questions from T2, T3, T4



What is a common repeater frequency offset in the 2-meter band?

- A. $\pm$ 500 kHz
- B. $\pm$ 600 kHz
- C. $\pm$ 1 MHz
- D. $\pm$ 5 MHz



What is the purpose of a CTCSS or PL tone?

- A. To boost transmit power
- B. To encrypt voice communications
- C. To reduce interference by activating the repeater only with the correct tone
- D. To identify your station automatically



Checkpoint Quiz

Q3

What is the national simplex calling frequency for FM in the 2-meter band?

- A. 144.000 MHz
- B. 146.520 MHz
- C. 146.940 MHz
- D. 148.000 MHz



What is the approximate wavelength of a 146 MHz signal?

- A. 1 meter
- B. 2 meters
- C. 6 meters
- D. 10 meters



Which propagation mode allows HF signals to travel long distances?

- A. Ground wave
- B. Line of sight
- C. Sky wave (ionospheric refraction)
- D. Tropospheric ducting



What can cause VHF signals to travel over 1,000 miles?

- A. D-layer absorption
- B. Sporadic E
- C. Ground wave
- D. Multipath



What organization manages ARES?

- A. FCC
- B. FEMA
- C. ARRL
- D. DHS



What does the squelch control do?

- A. Increases transmit power
- B. Adjusts the antenna tuner
- C. Mutes the receiver until a signal exceeds a set level
- D. Controls modulation percentage



When may you use any amateur frequency to call for help?

- A. During a net
- B. When propagation is poor
- C. At any time during a genuine emergency
- D. Only during declared RACES emergencies



What does a DMR color code do?

- A. Changes frequency offset
- B. Encrypts the signal
- C. Functions like a CTCSS tone for digital
- D. Sets the time slot



Before Next Session

Your assignment for Module 3



Your Assignment

- Drill HamStudy: T2A–T2C, T3A–T3C, T4A–T4B
- Target: 80%+ confidence on each section
- Coming in Module 3: Technical Core
 - Ohm's Law ($V = I \times R$) and power ($P = V \times I$)
 - Decibels (+3 dB = double power, that's basically it)
 - Resistors, capacitors, inductors, transistors
 - Reading schematics
- Don't panic — the math is 4 formulas and that's it
- See you next session!